



NARAYANA
IIT ACADEMY
INDIA



Sec: JR.IIT_*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 23-06-24
Max. Marks: 186

23-06-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_WAT-11_SYLLABUS

- MATHEMATICS:** **PRESENT WEEK (80%):** Conditional Identities, Summation of Trigonometric series, Miscellaneous problems on Transformation
PREVIOUS WEEK (20%): TRIGINOMETRY: Measurement of Angles, Trigonometric Ratios & Identities, T- Ratio of Compound angle, T-Ratio of Multiple, sub-multiple angles (Excluding graph, periodicity and extreme values), T-Ratio of some special angles, Transformation formulae
(Except Summation)
- PHYSICS:** **PRESENT WEEK (80%): FRICTION:** Contact Force and its components, Dry Friction, Static and Dynamic Friction, Laws of Friction, Angle of Friction & Repose, Rolling Friction
PREVIOUS WEEK (20%): Springs, Cutting & Combination of Springs, constraint equations, Problems in NLM without friction
- CHEMISTRY:** **PRESENT WEEK (80%):** Buffer solution & Buffer capacity, Solubility of sparingly soluble salts and solubility product, Indicator, Acid base titration curves
PREVIOUS WEEK (20%): Ionic Equilibrium-I: Theories of acid & bases – Arhenius, Bronsted-Lowry and Lewis theory **(Exclude: Comparision of Acidic & Basic strength)**, Ionic product of water and pH scale pH calculation involving strong acid, Strong base and their mixtures, pH of weak acids and bases, Mixture of weak acids, Levelling effect, Common ion effect, Dissociation of polyprotic acids, Salts and their Hydrolysis

PHYSICS:

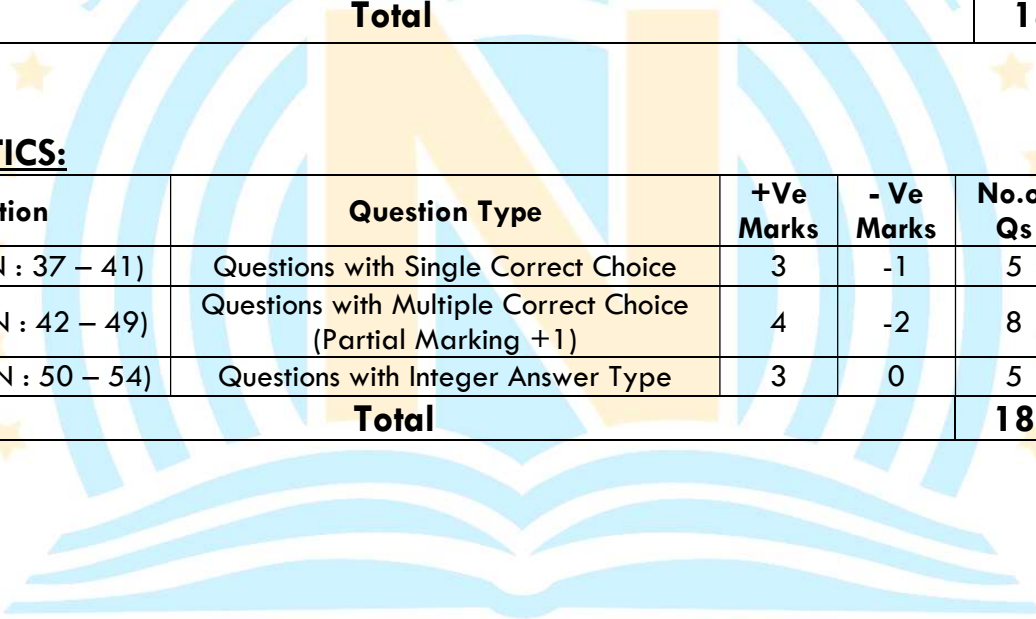
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 5)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 6 – 13)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 14 – 18)	Questions with Integer Answer Type	3	0	5	15
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 23)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 24 – 31)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 32 – 36)	Questions with Integer Answer Type	3	0	5	15
Total				18	62

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 41)	Questions with Single Correct Choice	3	-1	5	15
Sec – II(Q.N : 42 – 49)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 50 – 54)	Questions with Integer Answer Type	3	0	5	15
Total				18	62



THE NARAYANA GROUP

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. If $\sin x + \cos y = p$, $\cos x + \sin y = q$, then $\tan\left(\frac{x-y}{2}\right) =$

A) $p+q$ B) $p-q$ C) $\frac{p+q}{p-q}$ D) $\frac{p-q}{p+q}$
2. If $45^\circ < x < 90^\circ$ and $\sin x + \cos x = \frac{\sqrt{7}}{2}$, then $\cos^3 x - \sin^3 x =$

A) $\frac{11}{16}$ B) $-\frac{11}{16}$ C) $\frac{9}{16}$ D) $-\frac{9}{16}$
3. In a $\triangle ABC$, $\frac{\sin A}{\sin B} = 2 + \sqrt{3}$ & $\angle C = 60^\circ$ then $A + 2B =$

A) 135° B) 180° C) 150° D) 140°
4. If $\cos A + \cos B + \cos C = 0$, then value of k so that $\cos 3A + \cos 3B + \cos 3C = k \cos A \cos B \cos C$

A) 3 B) 12 C) 18 D) 27
5. If $2 \cos \alpha + 2 \cos \beta - 2 \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta = 3$, where $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, then $\alpha + \beta =$

A) $\frac{\pi}{3}$ B) $\frac{2\pi}{3}$ C) $\frac{\pi}{2}$ D) $\frac{5\pi}{6}$

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

6. Let $f(x, y) = \cos^2 x + \cos^2 y - \cos x \cos y$, then

A) $f(x, y) = \frac{3}{4}$ when $x + y = 60^\circ$ B) $f(x, y) = \frac{1}{4}$ when $x + y = 60^\circ$

C) $f(x, y) = \frac{3}{4}$ when $x - y = 60^\circ$ D) $f(x, y) = \frac{1}{2}$ when $x - y = 60^\circ$

7. If $\frac{\cos(\alpha - 3\theta)}{\cos^3 \theta} = \frac{\sin(\alpha - 3\theta)}{\sin^3 \theta} = m$ and if $\cos \alpha = f(m)$, then
- A) $f(\sqrt{2}) = 0$ B) $f(-2) = -1$ C) $f(\sqrt{3}) = -\frac{1}{\sqrt{3}}$ D) $f(-2) = 1$
8. If $p = \sin(A - B)\sin(C - D)$, $q = \sin(B - C)\sin(A - D)$, $r = \sin(C - A)\sin(B - D)$.
Then, which of the following is/are always true?
- A) $p + q + r = pqr$ B) $p^2 + q^2 + r^2 = pq + qr + rp$
C) $p + q + r = 0$ D) $p^3 + q^3 + r^3 - 3pqr = 0$
9. If $f_n(\theta) = \sum_{r=1}^n \tan\left(\frac{\theta}{2^r}\right) \sec\left(\frac{\theta}{2^{r-1}}\right)$, then which of the following is/are CORRECT?
- A) $f_6(48\pi) = 1$ B) $f_5(4\pi) = \sqrt{2} - 1$
C) $f_4\left(\frac{4\pi}{3}\right) = 2(\sqrt{3} - 1)$ D) $f_3(2\pi) = -1$
10. If $f_n(x) = \prod_{r=0}^{n-1} (2 \cos(2^r x) - 1)$, $n \geq 1$; then which of the following is/are INCORRECT?
- A) $f_5\left(\frac{2\pi}{33}\right) = -1$ B) $f_5\left(\frac{6\pi}{255}\right) = 1$ C) $f_5\left(\frac{2\pi}{31}\right) = 1$ D) $f_5\left(\frac{4\pi}{127}\right) = 1$
11. If $\cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + \beta + \gamma) = 0$ and $\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + \beta + \gamma) = 0$, where $0 < \beta, \gamma < \pi$, then
- A) $\beta + \gamma = \frac{4\pi}{3}$ B) $\beta + \gamma = \frac{2\pi}{3}$ C) $\beta - \gamma = 0$ D) $\beta - \gamma = \frac{\pi}{3}$
12. If $x \cos \alpha + y \sin \alpha = x \cos \beta + y \sin \beta = 2a$ ($0 < \alpha, \beta < \pi/2$) then
- A) $\cos \alpha + \cos \beta = \frac{4ax}{x^2 + y^2}$ B) $\cos \alpha \cdot \cos \beta = \frac{4a^2 - y^2}{x^2 + y^2}$
C) $\sin \alpha + \sin \beta = \frac{4ay}{x^2 + y^2}$ D) $\sin \alpha \cdot \sin \beta = \frac{4a^2 - y^2}{x^2 + y^2}$

13. If $A + B + C + D = 2\pi$, then $\cos A + \cos B + \cos C + \cos D =$

A) $4 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{B+C}{2}\right) \cos\left(\frac{C+A}{2}\right)$

B) $4 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{B+C}{2}\right) \cos\left(\frac{C+D}{2}\right)$

C) $4 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{B+D}{2}\right) \cos\left(\frac{D+A}{2}\right)$

D) $4 \cos\left(\frac{A-B}{2}\right) \cos\left(\frac{B-C}{2}\right) \cos\left(\frac{C+D}{2}\right)$

SECTION – III
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

14. The value of $\frac{1}{2} \left(\frac{1}{\cos^2 10^\circ} + \frac{1}{\sin^2 20^\circ} + \frac{1}{\sin^2 40^\circ} - \frac{1}{\cos^2 45^\circ} \right)$ is

15. $\operatorname{cosec} 2^\circ + \operatorname{cosec} 4^\circ + \operatorname{cosec} 8^\circ + \cot 8^\circ = \cot \alpha^\circ, \alpha \in [0, 10)$, then the value of α is...

16. If $\sum_{r=1}^n \frac{\cos(2rx)}{\sin\left(rx + \frac{\pi}{4}\right)} = \frac{-2 \sin(50x) \sin\left(\frac{\pi}{4} - \frac{101x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$, then value of $\left[\frac{n}{12}\right] =$

17. If $\cos A = \frac{3}{4}$, then $\frac{32}{11} \sin \frac{A}{2} \sin \frac{5A}{2} =$

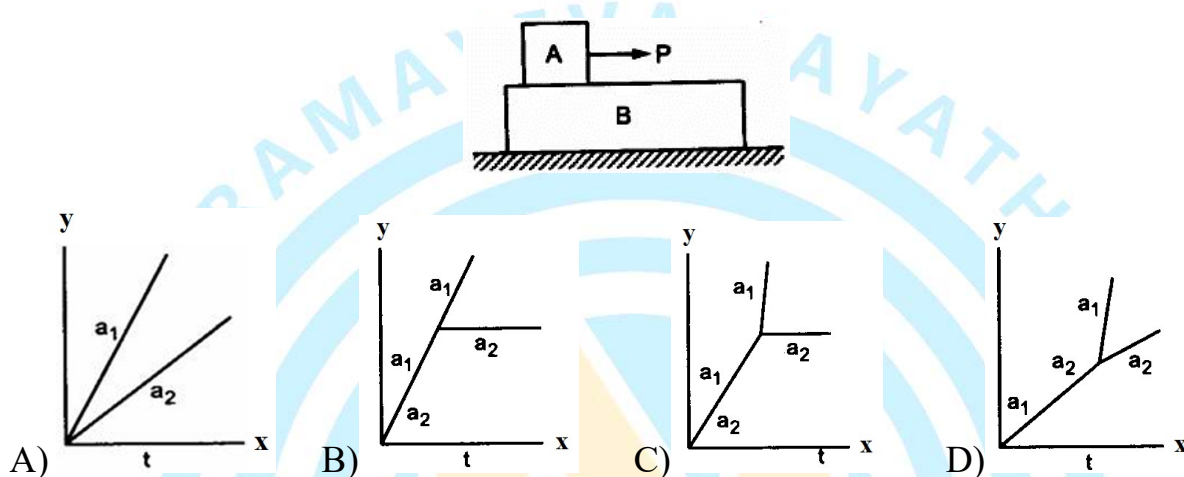
18. $\sqrt{7} \left(\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \right)^{-1} =$

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

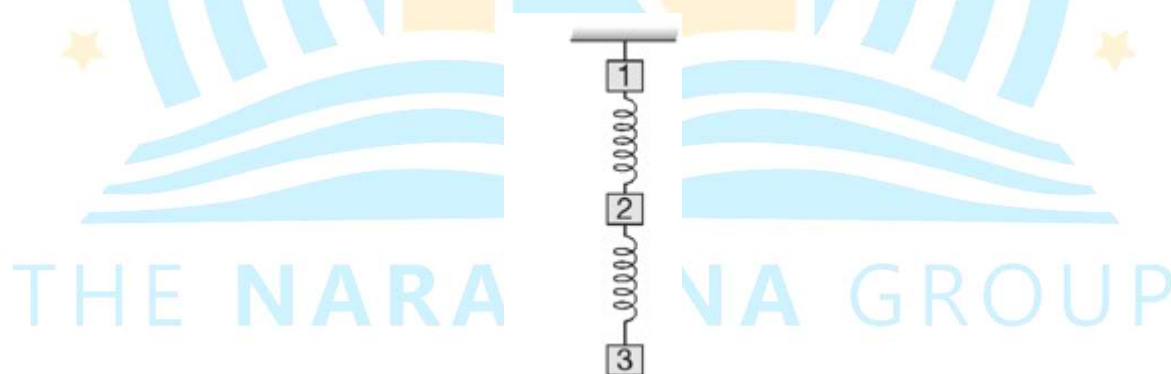
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

19. Block A is placed on B, whose mass is greater than that of A. There is friction between the blocks, while the ground is smooth. A horizontal force P , increasing linearly with time, begins to act on A. The acceleration a_1 and a_2 of A and B respectively are plotted against time (t). Choose the correct graph

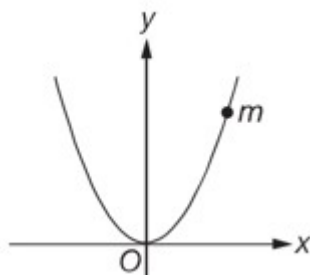


20. Three identical blocks are suspended on two identical springs one below the other as shown in figure. If thread is cut that supports block 1, then initially (choose one alternative only)

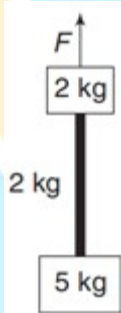


- A) The second block falls with zero acceleration
B) The first block falls with maximum acceleration
C) Both (A) and (B) are wrong
D) Both (A) and (B) are correct

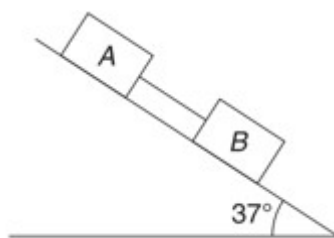
21. A bead of mass m is located on a parabolic wire with its axis vertical directed towards downward as in figure and whose equation is $x^2 = ay$. If the coefficient of friction is μ , the highest distance above the x – axis at which the particle will be in equilibrium is



- A) μa B) $\mu^2 a$ C) $\frac{1}{4} \mu^2 a$ D) $\frac{1}{2} \mu a$
22. In the figure shown, the uniform vertical rope joining the two blocks has a mass of 2kg. A constant vertical force of $F=180$ N is acting on the upper block and pulling the system upwards with some acceleration. Then



- A) Tension at the midpoint of the rope is 55 N
 B) The rope pulls the 2kg block with larger force than the force it exerts on 5kg block.
 C) Acceleration of the 5 kg block is 2 ms^{-2}
 D) None of the above
23. Two blocks A and B, each of mass 10 kg, are sliding on an incline. They are connected by a string. The coefficients of kinetic friction between A and incline is $\mu_1 = 0.5$ and that between B and incline is $\mu_2 = 0.4$. The tension in string will be



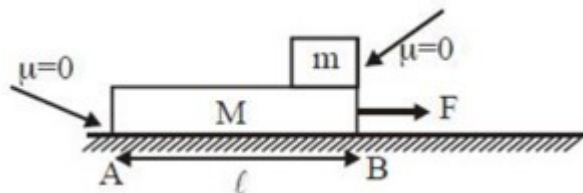
- A) Zero B) 2 N C) 8 N D) 4 N

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

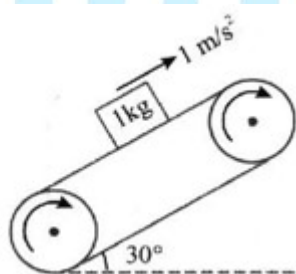
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

24. In the figure small block of mass m is kept on M then

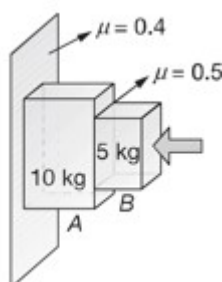


- A) The acceleration of m w.r.t. ground is $\frac{F}{m}$
- B) The acceleration of m w.r.t. ground is zero
- C) The time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$
- D) The time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

25. A block of mass 1 kg is stationary with respect to a conveyer belt that is accelerating with 1 m/s^2 upwards at an angle of 30° as shown in figure. The true statements of the following is / are

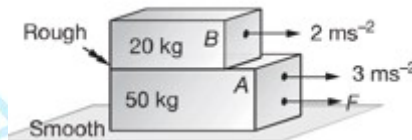


- A) Force of friction on block is 6N upalong the belt.
- B) Force of friction on block is 1.5 N up the belt.
- C) Contact force between the block and belt is 10.5 N(nearly)
- D) Contact force between the block and belt is $5\sqrt{3} \text{ N}$.
26. Two blocks are pushed against a rough wall and the system is at rest as shown in figure. Select the correct statements (s).

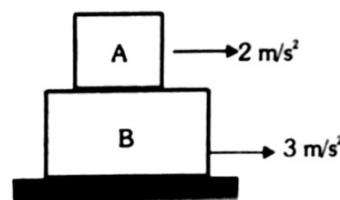


- A) Friction force on block A due to B is upward direction
- B) Friction force on block A due to wall is 150 N
- C) Minimum force required to keep system at rest is 375 N
- D) Friction of block B due to A increases with increase in force applied F

27. A 20 kg block is placed on top of 50 kg block as shown in figure. A horizontal force F acting on A produces an acceleration of 3ms^{-2} in A and 2ms^{-2} in B as shown in figure. For this situation mark out the correct statement(s).



- A) The magnitude of friction force between A and B is 40 N.
 - B) The magnitude of net force acting on A is 150 N.
 - C) The magnitude of F is 190 N.
 - D) The magnitude of F is 150 N
28. Block A is kept on block B as shown in figure. It is known that acceleration of block A is 2 m/s^2 towards right and acceleration of block B is 3 m/s^2 towards right under the effect of unknown forces. The friction force acting on A by B ($\mu_{AB} = 0.3$)

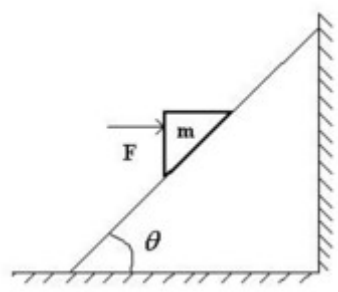


- A) Is necessarily towards right
 - B) May be towards right
 - C) May be towards left
 - D) May be zero
29. A force F_1 that is necessary to move a body (Weight w) up an inclined plane is double the force F_2 that is necessary to just prevent it from sliding down. F_1 and F_2 are parallel to the incline. Which of these options are correct?

Where ϕ := angle of friction, θ = angle of inclined plane and w = weight of the body

- A) $F_2 = w \sin(\theta - \phi) \sec \phi$
- B) $F_1 = w \sin(\theta - \phi) \sec \phi$
- C) $\tan \phi = 3 \tan \theta$
- D) $\tan \theta = 3 \tan \phi$

30. Which of the following statement(s) is (are) correct about friction?
- A) The coefficient of friction between two bodies is largely independent of area of contact.
 - B) The frictional force can never exceed the reaction force on a body from the supporting surface.
 - C) friction is purely gravitational force
 - D) Friction is due to irregularities of the surfaces in contact
31. A triangular block of mass m rests on a fixed rough inclined plane having friction coefficient μ with the block. A horizontal force F is applied to it as shown in figure below, then the correct statement is



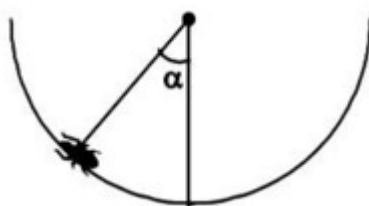
- A) Friction force is zero when $F \cos \theta = mg \sin \theta$
- B) The value of limiting friction is $\mu(mg \sin \theta + F \cos \theta)$
- C) Normal reaction on the block is $F \sin \theta + mg \cos \theta$
- D) The value of limiting friction is $\mu(mg \sin \theta - F \cos \theta)$

SECTION – III
(INTEGER ANSWER TYPE)

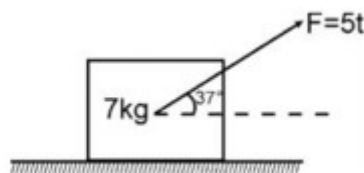
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Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

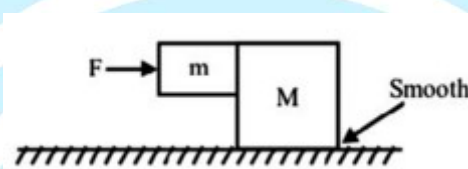
32. An insect crawls up a hemispherical surface very slowly (see fig). The coefficient of friction between the insect and the surface is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, what is the minimum possible value of $\cot \alpha$?



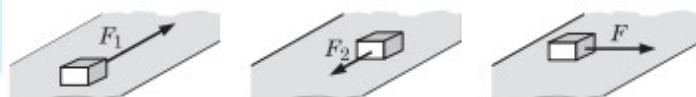
33. A block of 7kg is placed on a rough horizontal surface and is pulled through a variable force F (in N) = $5t$ where 't' is time in second at an angle of 37° with the horizontal as shown in figure. The coefficient of static friction of the block with the surface is one. If the force starts acting at $t = 0$ sec then the block starts to slide at time $t = 10(x)$ sec. Find x. [Take $g = 10\text{ms}^{-2}$].



34. The two blocks, $m = 10$ kg and $M = 50$ kg are free to move as shown. The coefficient of static friction between the blocks is 0.5 and there is no friction between M and the ground. What is minimum horizontal force F required to hold m against M (in newton)?

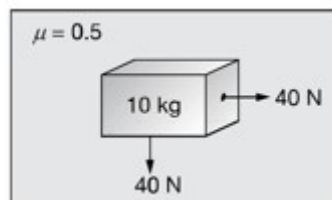


35. To drag a block up an inclined plane with a constant speed, a force $F_1 = 8$ N has to be applied along the line of fastest descent, whereas to drag the block down the plane with a constant speed, a force $F_2 = 2$ N has to be applied along the line of fastest descent as shown in the following figure. Find the minimum horizontal force F parallel to the plane required to slide the block (in Newton)



36. A block of mass 10 kg is placed on a horizontal rough table surface. The top view of the block on the table is shown. the acceleration of the block, (taking $\mu = 0.5$ and

$g = 10\text{ms}^{-2}$.) is $\frac{n}{10} \text{ms}^{-2}$. Find n



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. The solubility of AgCl is maximum in

- A) 0.1M NaCl B) 0.1M HCl C) 55.5 M H_2O D) 0.1M $AgNO_3$

38. A solution is 0.10 M in Ag^+ , Ca^{2+} , Mg^{2+} and Al^{3+} ions. Which compound will precipitate at the lowest $[PO_4^{3-}]$ when a solution of Na_3PO_4 is added?

- A) Ag_3PO_4 ($K_{sp} = 1 \times 10^{-16}$) B) $Ca_3(PO_4)_2$ ($K_{sp} = 1 \times 10^{-33}$)
C) $Mg_3(PO_4)_2$ ($K_{sp} = 1 \times 10^{-24}$) D) $AlPO_4$ ($K_{sp} = 1 \times 10^{-20}$)

39. A solution is 0.1 M in Cl^- ion, 0.1 M in Br^- ion and 0.1 M in I^- ion. Solid $AgNO_3$ is gradually added to this solution. Assume the addition of $AgNO_3$ does not change the volume. The concentration of I^- and Br^- ions in solution when Cl^- ion starts precipitating are respectively, Given $K_s(AgCl) = 1.7 \times 10^{-10} M^2$; $K_s(AgBr) = 5.0 \times 10^{-13} M^2$; $K_s(AgI) = 8.5 \times 10^{-17} M^2$

The concentration of I^- and Br^- ions in solution when Cl^- ion starts precipitating are respectively,

- A) $5 \times 10^{-8} M$ and $3.0 \times 10^{-4} M$ B) $3 \times 10^{-4} M$ and $5 \times 10^{-8} M$
C) $3 \times 10^{-4} M$ and $8 \times 10^{-8} M$ D) $5 \times 10^{-8} M$ and $3.0 \times 10^{-5} M$

40. The acid dissociation constant (K_a) for $HCOOH$, CH_3COOH , $CH_2ClCOOH$ and HCN at $25^\circ C$ are 1.8×10^{-4} , 1.8×10^{-5} , 1.4×10^{-3} and 4.8×10^{-10} , respectively. The acid that gives highest pH at the equivalence point when 0.2 M solution of each acid is titrated with a 0.2 M solution of sodium hydroxide is:

- A) $HCOOH$ B) CH_3COOH C) $CH_2ClCOOH$ D) HCN

41. Choose the correct code

Column – I		Column – II	
(P)	pK_b of X^- (K_a of $HX = 10^{-6}$)	(1)	6.79
(Q)	pH of 10^{-7} M HCl at 25°C ($\log = 0.3$)	(2)	8
(R)	pH of 10^{-2} M acetic solution at 25°C (K_a of acetic acid $= 1.6 \times 10^{-5}$; $\alpha \ll 1$)	(3)	4.3
(S)	pH of the mixture obtained by mixing equal volumes of acids with pH 4 and 5.	(4)	3.4

Codes:

	P	Q	R	S
A)	1	2	4	3
B)	4	3	2	1
C)	2	1	4	3
D)	1	2	3	4

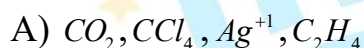
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

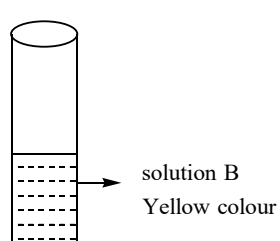
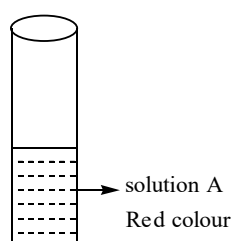
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Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

42. Which of the following set of species can act as Lewis acids

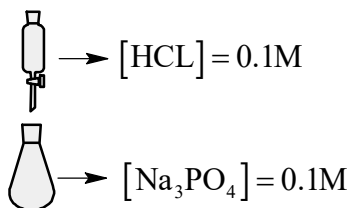


43. The indicator methyl orange (pH range = 3.1 to 4.5) has been added to both of the following solutions at 25°C . Based on the colours, which of the following statements are correct:



- A) The pH of solution A is definitely less than 7.
 B) The pH of solution B is definitely greater than 7.
 C) The pH of solution B is greater than pH of solution A.
 D) The pH of solution A is greater than pH of solution B.
44. Which can act as buffer?
 A) 200 ml of 0.5M $HCOOH$ + 100 ml of 0.4M KOH
 B) 0.1 mole of CH_3NH_2 + 0.08 moles of HCl
 C) 40 ml of 0.1M NaCN + 20 ml of 0.1M HCl
 D) 50 ml of 0.2M CH_3COOH + 25 ml of 0.4M NH_4OH
45. A 2.5 g impure sample containing weak monoacidic base (molecular weight = 45) is dissolved in 100 ml water and titrated with 0.5 M HCl at $25^\circ C$. When $\left(\frac{1}{5}\right)^{th}$ of the base was neutralized, the pH was found to be 9 and at equivalent point, the pH of solution is 4.5. Which of the following is/are correct option(s)?
 [Given: $\log 2 = 0.3$; $\log 2.5 = 0.4$]
 A) K_b of weak monoacidic base is less than 2.5×10^{-6} .
 B) Concentration of salt at equivalence point is 0.25 M.
 C) Volume of HCl used at equivalence point is 100 ml.
 D) Mass percentage of weak base in given sample is 90%.
46. The aqueous solutions of which of the following gives pink colour with phenolphthalein
 A) Na_2CO_3 B) $CuSO_4$ C) $NaCN$ D) CH_3COONa
47. Which of the following statements are **incorrect** ?
 A) The concentration of Fe^{3+} will increase in a solution of $Fe(OH)_3$ when $[H^+]$ decrease (Given K_{sp} of $Fe(OH)_3$ is 4×10^{-18} at 298 K).
 B) In a mixture of NH_4Cl and NH_4OH in water, when a further amount of NH_4Cl is added, the pH of mixture will decrease.
 C) When HCl dissolve in benzene, HCl act as Bronsted acid.
 D) When 100ml of 1M CH_3COOH is added with 900 ml of water its pH decreases (at same temperature).

48. Titration of Na_3PO_4 solution by HCl solution



$V = 50 \text{ ml}$

Where K_{a1} , K_{a2} and K_{a3} are dissociation constants of H_3PO_4

($pK_{a1} = 4$; $pK_{a2} = 8$; $pK_{a3} = 12$; $\log 2 = 0.3$; $\log 3 = 0.48$; $\log 4 = 0.6$)

- A) When 10 ml of HCl is added, the pH of resulting solution is 12.6.
- B) When 50 ml of HCl is added, the pH of resulting solution is 10.
- C) When 100 ml of HCl is added, the pH of resulting solution is 6.
- D) When 120 ml of HCl is added, the pH of resulting solution is 4.18.
49. Which of the following are correct ?
- A) During titration between Na_2CO_3 verses HCl in presence of phenolphthalein, end point is reached at half neutralization of Na_2CO_3
- B) Blood is an acidic Buffer
- C) In the titration of Acetic acid solution with Sodium hydroxide solution, Methyl Orange is used as suitable indicator.
- D) In the titration of Hydrochloric acid solution with Sodium hydroxide solution, only Methyl Orange is used as suitable indicator and any other is not suitable indicator.

SECTION – III (INTEGER ANSWER TYPE)

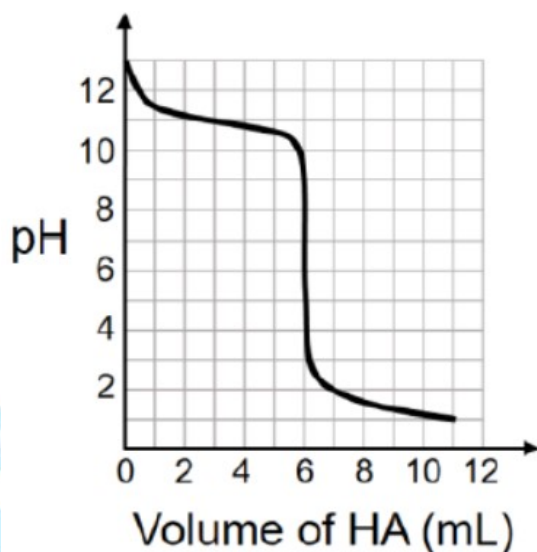
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

50. A solution is prepared by mixing 0.01 mol each of H_2CO_3 , NaHCO_3 , Na_2CO_3 , and NaOH in 100 mL of water. pH of the resulting solution is $(2 \times y)$. The value of 'y' is__

[Given: pK_{a1} and pK_{a2} of H_2CO_3 are 6.37 and 10.3, respectively; $\log 2 = 0.30$]

51. A solution of 0.1 M weak base (B) is titrated with 0.1 M strong acid (HA). The variation of pH of the solution with the volume of HA added is shown in the figure below. What is the pK_b of the base? The neutralization reaction is given by $B + HA \rightarrow BH^+ + A^-$.



52. Concentration of H_2SO_4 and Na_2SO_4 in a solution is 1 M and 1.8×10^{-2} M respectively. Molar solubility of $PbSO_4$ in the same solution is $X \times 10^{-Y}$ M (expressed in scientific notation). The value of Y is _____.
[Given: Solubility product of $PbSO_4$ (K_{sp}) = 1.6×10^{-8} . For H_2SO_4 , K_{a1} is very large and $K_{a2} = 1.2 \times 10^{-2}$]
53. K_a for HCN is 5×10^{-10} at $25^\circ C$. For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 mL of 2 M HCN solution is
54. The Molar solubility of $Cr(OH)_3$ in a solution where $[H^+]$ is 100 times $[OH^-]$ is $(x \times 10^{-y}) \text{ mole.lit}^{-1}$ in scientific notation at $25^\circ C$. Find the value of $(y - x)$ if K_{sp} of $Cr(OH)_3$ is 6×10^{-32} .